xengsort2: Ultrafast accurate xenograft sorting

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Introduction

With an increasing number of patient-derived xenograft (PDX) models being created and subsequently sequenced to study tumor heterogeneity and to guide therapy decisions, there is a need for methods to **separate reads** originating from the **graft** (human) tumor and reads originating from the **host** species' (mouse) surrounding tissue.

Two kinds of methods are in use:

- alignment-**based** tools like **XenofilteR** [1] require that reads are mapped and aligned to the host and graft genomes and then processes the resulting alignments and quality metrics
- alignment-**free** tools like **xengsort** [2] work directly on the raw read data

We here argue that a carefully engineered alignment-free approach using subdivided three-way bucketed Cuckoo hashing [3, 4] offers high read classification accuracy and a fast running time.

Problem

Given: reads from **mouse** and **tumor**

Goal: assign each read to one of 5 categories:

- host (mouse)
- graft (human)
- both
- ambiguous
- neither

Basic Idea

- Build an index of reference genomes using a subdivided 3-way bucketed Cuckoo hash table
  - Divide reference genomes into (gapped) *k*-mers
  - Store each ***k*-mer** as a **key** and the corresponding **species** as a **value** in the index
  - If an *k*-mer occurs in both references, store both
  - Calculate Hamming distance between all all stored (gapped) *k*-mers
  - Mark *k*-mers from different species with Hamming distance one as **weak**
- Classify each read using its *k*-mers
  - For each *k*-mer, query the corresponding species in the index
  - Assign category based on values using specific rules

New Features

- Support for **gapped** *k*-mers
- Config files for different settings (short and long reads)
- **Parallelized** index building
- Improved weak *k*-mer algorithm
- Compressed In-/Output (`.gz`, `.bz2`, `.xz`)
- New **coverage based** classification algorithm
- 1000s of samples: load hash table as **shared memory**

References

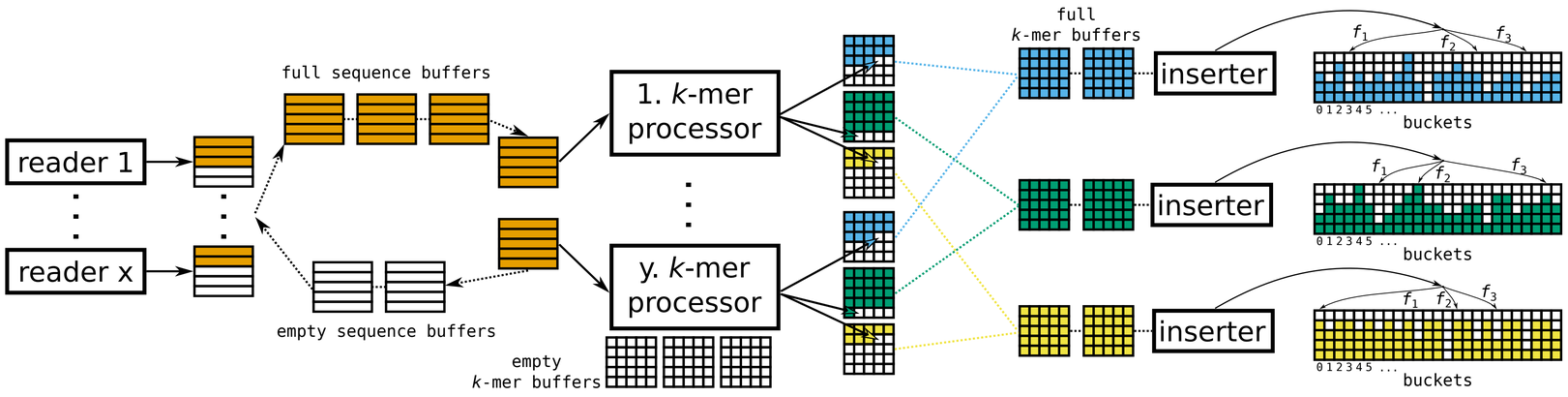
[1] R. J. C. Kluin et al. XenofilteR: computational deconvolution of mouse and human reads in tumor xenograft sequence data. *BMC Bioinformatics*, 19(1):366, Oct 2018.

[2] Jens Zentgraf and Sven Rahmann. Fast lightweight accurate xenograft sorting. In *20th International Workshop on Algorithms in Bioinformatics (WABI 2020)*. Schloss Dagstuhl-Leibniz-Zentrum fuer Informatik, 2020.

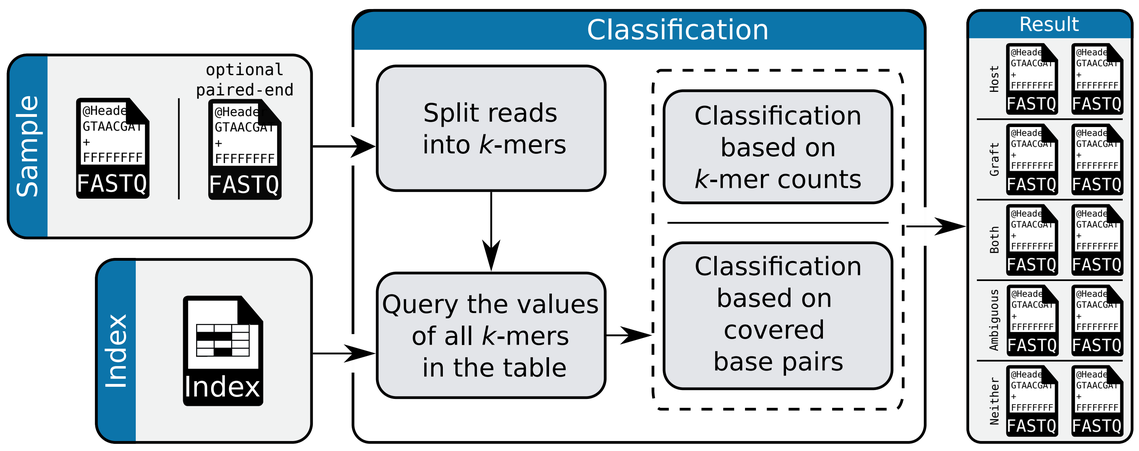
[3] J. Zentgraf et al. Cost-optimal assignment of elements in genome-scale multi-way bucketed cuckoo hash tables. In *Proceedings of the Symposium on Algorithm Engineering and Experiments (ALENEX) 2020*, pages 186–198. SIAM, 2020.

[4] Jens Zentgraf and Sven Rahmann. Fast gapped k-mer counting with subdivided multi-way bucketed cuckoo hash tables. In *22nd International Workshop on Algorithms in Bioinformatics (WABI 2022)*, 2022.

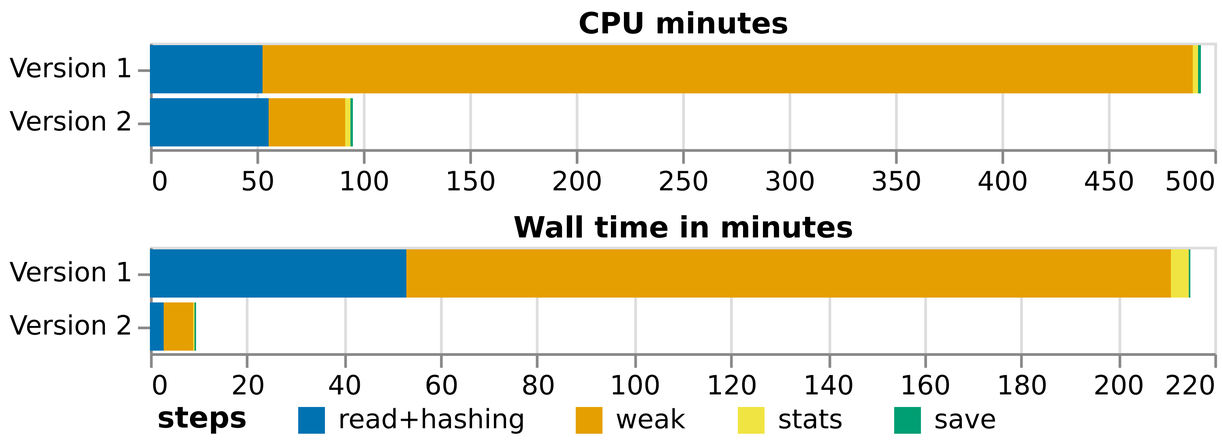
Index building steps



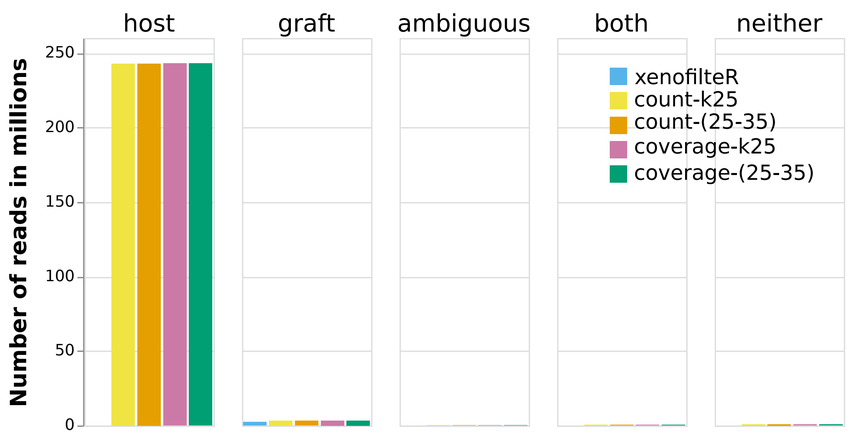
Classification workflow



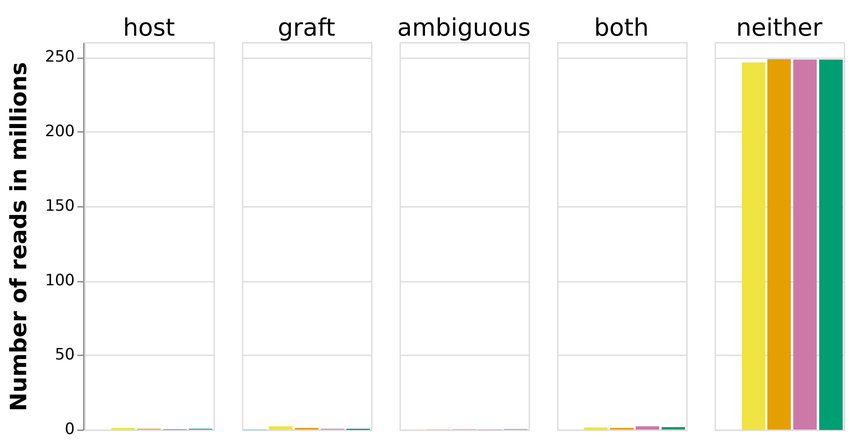
Index build time



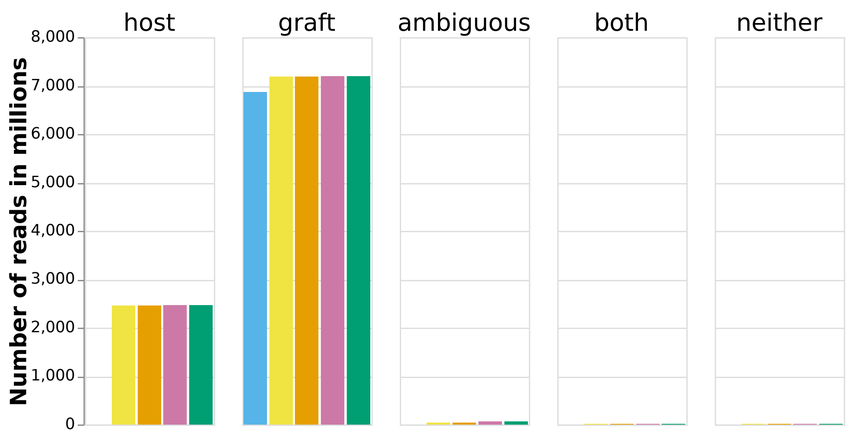
Mouse Exome



Chicken Genome



PDX RNA



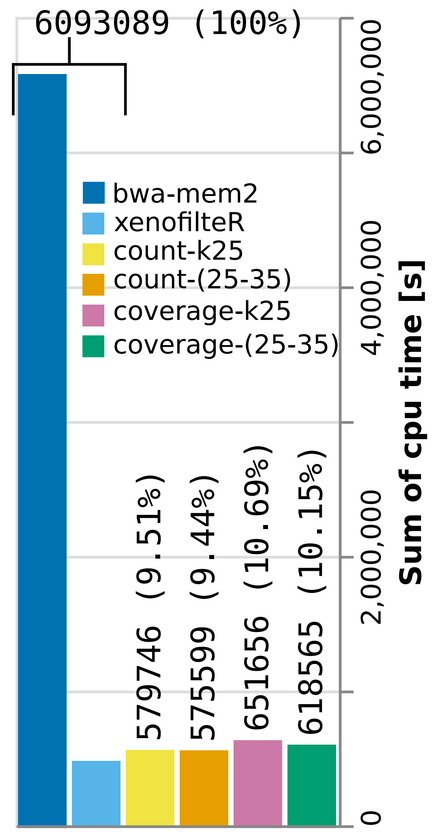
Datasets for Experiments

name	size (M)
mouse exomes captured	
with a human exome capture kit	307
chicken genome from a whole blood sample	251
PDX RNA from human pancreatic tumors	9742

Comparison between 2 different tools:

- **xenofilteR** [1] (alignment based)
- **xengsort** [2] (*k*-mer based)
- Each tool is assigned **8** threads per run

Masks:

- `##########################`
- `#####_#_##_###__###__###_##_#_#####`

Run Time on PDX RNA